Finite thermodynamic reconstruction of an integer-spectrum CFT

For a compact unitary two-dimensional CFT with $c_L = c_R = 12k$, integer total dimensions and spins, and unmultiplied modular invariance, finitely many exact angular moments at finitely many temperatures determine the torus partition function. We prove explicit sampling bounds using separate $\Gamma(2)$ invariance. The angular order is optimal: for every k we construct two genuine Narain CFTs with identical lower moments at every temperature but different moments at the stated order. The result determines the torus spectrum, not operator products; it assumes exact data and does not establish numerical stability.

Thermal data. The torus trace encodes energy and angular momentum, but its zero-angle restriction sums over spin. Modular spectral reconstruction and discreteness have substantial literature [1–3]. Integer-energy partition functions and their thermal chiralization were studied by Ashrafi and Loran [1]. Dymarsky and Shapere exhibited Narain CFTs with identical thermal traces but distinct angular traces [4]; Kaidi and Perlmutter discussed earlier CFT pairs with the same light spectrum [2]. Benjamin *et al.* studied reconstruction from light, scalar and spin-one spectra [3], assuming a simple cuspidal eigenspectrum. We give a finite-sampling consequence of integer dimensions and full modular invariance.

Fix the spatial circle to circumference 2π and write

$$\begin{aligned} Z(z+it,-z+it) &= \operatorname{Tr} e^{-2\pi t(\Delta-k)+2\pi i z J}, \\ M_r(t) &= \operatorname{Tr}[J^r e^{-2\pi t(\Delta-k)}]. \end{aligned} \tag{1}$$

Here $t = \beta/(2\pi) > 0$ is the rescaled inverse temperature and the moments include descendants. Assume a compact unitary bosonic CFT with discrete finite-multiplicity spectrum, a unique vacuum, integer $\Delta = h + \bar{h}$ and $J = h - \bar{h}$, $c_L = c_R = 12k$ for $k \in \mathbb{Z}_{>0}$, and simultaneous unmultiplied S, T invariance. Neither parity invariance nor rationality is assumed.

Theorem. Two theories satisfying these hypotheses for the same k have identical torus partition functions if their M_r agree for every $0 \le r \le R_k$ at either k distinct inverse temperatures all in $(0, 1]$, or all in $[1, \infty)$, or $2k-1$ arbitrary distinct positive inverse temperatures, where

$$R_k = \begin{cases} 3k, & k \text{ even}, \\ 3k-1, & k \text{ odd}. \end{cases} \tag{2}$$

These data are exact; equality determines spin-resolved state multiplicities, not OPE coefficients. Alternatively choose k distinct $t > 1$, use all for $r = 0$, and use any $k - \lceil r/3 \rceil + 1$ of them at each $r \ge 1$.

Proof. Continue the trace holomorphically to independent τ, σ in the upper half-plane, with physical slice $\sigma = -\bar{\tau}$. Normal convergence and holomorphic uniqueness on this maximally totally real slice continue simultaneous modular invariance. Since $2h, 2\bar{h} \in \mathbb{Z}$, shifting either variable by 2 has no phase: $e^{4\pi i(h-k/2)} = 1$. Conjugating independent T^2 by simultaneous S gives separate $\Gamma(2)$ invariance.

For fixed σ, the left infinity cusp has pole order at most k in the width-two coordinate $u = e^{\pi i \tau}$. At the other cusps, simultaneous modular invariance moves that cusp to infinity and sends σ to another fixed point of the upper half-plane; the same absolute trace bound applies. The right-variable statement follows identically. Since $X(2)$ is the λ-sphere with cusps $0, 1, \infty$, each trace has

$$Z(\tau,\sigma) = \frac{P(x,y)}{[x(1-x)y(1-y)]^k}, \quad x = \lambda(\tau),\ y = \lambda(\sigma), \tag{3}$$

with $\deg_x P, \deg_y P \le 3k$. This genus-zero statement does not assume rationality; see also the level-two discussion in [5]. To obtain the two-variable form, evaluate at $3k+1$ independent first-variable points: the resulting coefficient functions inherit the right cusp bounds as finite linear combinations of traces.

Let P be the difference of the two numerators and put $n = 3k$. The map $t \mapsto \lambda(it)$ is strictly monotone from 1 to 0, and

$$\lambda'(\tau) = \pi i \lambda(\tau)(1-\lambda(\tau))\vartheta_3(\tau)^4 \neq 0 \quad (\tau = it). \tag{4}$$

Suppose inductively $P = (x-y)^r Q_r$. The rth angular derivative at $z = 0$ is a nonzero factor times $q_r(x) = Q_r(x,x)$, with $\deg q_r \le 2(n-r)$. Thus vanishing moments give zeros of q_r. Simultaneous modular transformations act on x, y by the same S_3 Möbius transformation. The denominator in (3) and $(x-y)^r$ acquire nonzero section factors, so a zero propagates through its orbit. A generic orbit has six finite points, exactly two in $(0,1)$, namely $x, 1-x$; the exceptional orbit is $\{1/2, -1, 2\}$. The samples meet at least k orbits and give at least $6k-3$ distinct zeros, exceeding $2(n-r)$ for $r \ge 2$.

For $r = 0, 1$ use the common vacuum. Near the diagonal infinity cusp, the vacuum cancels and the remaining terms have total degree at least two in (u, v), by integer Δ and unitarity. Thus P has joint order at least two at every diagonal cusp, the other two following from modular covariance. As a section, q_0 has two additional zeros at each cusp and q_1 has one at each after division by $x - y$; a zero at infinity is a deficit from the maximum polynomial degree. Together with the orbit zeros, this forces $q_r = 0$ for $r = 0, 1$ too. Thus $x - y$ divides Q_r and the induction continues.

For even k, data through $r = 3k$ force $(x-y)^{3k+1} \mid P$, impossible by bidegree unless $P = 0$. For odd k, data

through $r = 3k-1$ leave only $P = C(x-y)^{3k}$; simultaneous S sends x, y to $1-x, 1-y$, fixing the denominator and reversing this odd polynomial, so $C = 0$. For reduced sampling every strictly one-sided sample has a distinct six-point orbit; the cusp zeros suffice at $r = 0$, and for $r \geq 1$ the stated count N_r obeys $6N_r > 6k - 2r$.

Low-energy corollary. The complete spin-resolved state spectrum through $N = \lfloor 3k/2 \rfloor$ determines Z. Indeed, for a difference with this jet zero, the cleared numerator has order at least $2(N+1)$ at each diagonal cusp. After r divisions by $x-y$, its diagonal restriction has degree $6k-2r$ and at least $3[2(N+1)-r]$ cusp zeros. Their number exceeds the degree for every $r \leq 3k$, forcing the difference to vanish. The physical pairs below first differ at $\Delta = N$, proving this cutoff sharp.

Physical seeds. Let U, V be the $D_{4,1}$ vacuum and vector characters; the spinor characters also equal V at zero chemical potentials. Define

$$\begin{aligned} A &= U(\tau)U(\sigma) + 3V(\tau)V(\sigma), \\ W &= U(\tau)V(\sigma) - V(\tau)U(\sigma). \end{aligned} \tag{5}$$

The $E_{8,1}$ character is $U^2 + 3V^2$. Hence the genuine $E_{8,1} \times D_{4,1}$ and $D_{4,1}^3$ theories have difference $3W^2A$. Their thermal equality follows from the D_4^2 versus E_8 example in [1], after tensoring with D_4. They have equal M_0, M_1 but unequal M_2 at every finite t.

For later composition use a neighboring pair instead: diagonal $D_{8,1} \times D_{4,1}$ and $D_{4,1}^3$. The D_8 diagonal characters give

$$Z_{D_8 \times D_4} - Z_{D_4^3} = W^2 A. \tag{6}$$

The D_8 root lattice is an index-two extension of D_4^2 by the vector–vector class. The corresponding maximal isotropic gluing codes therefore meet in codimension one. A concrete independent certificate is supplied with this source.

For a $k = 2$ neighboring pair, take the $D_4^6 \oplus D_4(-1)^6$ discriminant group $(\mathbb{F}_2^2)^{12}$. Consecutive bit pairs encode its twelve classes, the first six left-moving. Let v_i be

$$\begin{aligned} &(4280689, 4003343, 13490061, 15943387, \\ &4783610, 4517512, 8704049, 3143493, \\ &9304945, 11135812, 2974764, 11955798), \end{aligned}$$

indexed from $i = 0$. Set $a = v_1$, $a' = 5138$, $f = 30$, and let S be spanned by $v_i + \mathrm{bit}_i(f)v_1$ for $i \neq 1$. Then $S + \langle a \rangle$ and $S + \langle a' \rangle$ are neighboring maximal isotropic codes. The supplied standard-library certificate checks their 2^{11}-element intersection, both 2^{12}-element spans, the quadratic and bilinear forms, and the grouped weight enumerators. The latter give exactly

$$Z_{S+\langle a \rangle} - Z_{S+\langle a' \rangle} = W^6. \tag{7}$$

The initial lattice has determinant $4^{12} = 2^{24}$; adjoining either code gives index 2^{12} and an even unimodular lattice of signature $(24, 24)$. These define genuine compact unitary Narain CFTs [4, 6]. The vacuum is unique; isotropy gives integer spin and, since all three nonzero D_4 classes have norm 1 mod 2, integer total dimension. At the first changed level $\Delta = 3$, the sixth state-spin moments differ by $8^6 6! = 188{,}743{,}680$.

Sharpness for every k. We use the following elementary composition of neighboring isotropic codes. Write two pairs as $C = S + \langle a \rangle$, $C' = S + \langle a' \rangle$ and $D = T + \langle b \rangle$, $D' = T + \langle b' \rangle$, where the lifts are isotropic and the polarized pairings satisfy $(a, a') = (b, b') = 1$ in $\mathbb{F}_2$. In the orthogonal direct sum define

$$\begin{aligned} E_+ &= S \oplus T + \langle (a, b), (a', b') \rangle, \\ E_- &= S \oplus T + \langle (a, b'), (a', b) \rangle. \end{aligned} \tag{8}$$

The two added vectors are isotropic and mutually orthogonal because $1+1 = 0$. Both codes are maximal isotropic and they are neighbors: their common hyperplane is generated by $S \oplus T$ and $(a+a', b+b')$. Comparing their four cosets gives

$$Z_{E_+} - Z_{E_-} = (Z_C - Z_{C'})(Z_D - Z_{D'}). \tag{9}$$

Iterating (8) with the two physical seeds produces even unimodular Narain pairs whose partition-function differences are W^{3k} for even k and $W^{3k-1}A$ for odd k. They have equal moments through $R_k - 1$ for all t, while the R_kth differs. Indeed, the character identity $\lambda = 2V/(U+V)$ and (4) imply that W has a simple angular zero: $U(it) > 0$ and $\lambda'(it) \neq 0$. Also $A(it, it) > 0$. Thus (2) is physically sharp for every k. These pairs first differ at $\Delta = \lfloor 3k/2 \rfloor$, so the low-energy corollary is sharp as well. The certificate also explicitly checks the first composed $k = 3$ pair.

Scope. The theorem concerns exact torus data and makes no numerical stability claim. Positive integral modular character data alone do not supply sewing or crossing; examples are known [7]. Our sharpness pairs are actual lattice CFTs, not merely formal spectra. The $k = 2$ seed was found by a finite search and exactly certified. A complete literature and priority assessment remains open.

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